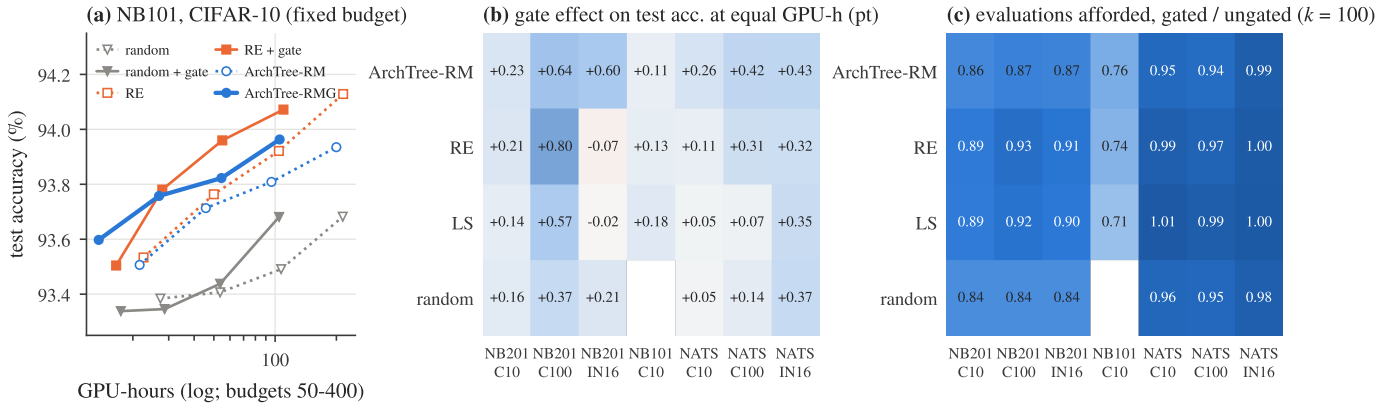

(a) NB101, CIFAR-10 (fixed budget)
random
random + gate
RE
RE + gate
ArchTree-RM
ArchTree-RMG
94.2
94.0
93.8
93.6
93.4
test accuracy (%)
100
GPU-hours (log; budgets 50-400)
(b) gate effect on test acc. at equal GPU-h (pt)
ArchTree-RM +0.23 +0.64 +0.60 +0.11 +0.26 +0.42 +0.43
RE +0.21 +0.80 -0.07 +0.13 +0.11 +0.31 +0.32
LS +0.14 +0.57 -0.02 +0.18 +0.05 +0.07 +0.35
random +0.16 +0.37 +0.21 +0.05 +0.14 +0.37
NB201 C10
NB201 C100
NB201 IN16
NB101 C10
NATS C10
NATS C100
NATS IN16
(c) evaluations afforded, gated / ungated (k = 100)
ArchTree-RM 0.86 0.87 0.87 0.76 0.95 0.94 0.99
RE 0.89 0.93 0.91 0.74 0.99 0.97 1.00
LS 0.89 0.92 0.90 0.71 1.01 0.99 1.00
random 0.84 0.84 0.84 0.96 0.95 0.98
NB201 C10
NB201 C100
NB201 IN16
NB101 C10
NATS C10
NATS C100
NATS IN16